

Pneumatic Mechanical Systems Lab
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1.0 Objective

The objective of this lab is to understand how pneumatic systems which involve thermal fluids are used to create translational motion within mechanical systems. Over the course of this lab a pneumatic system simulated in Simscape will be used in conjunction with an actuator to create a dynamic mechanical system response

2.0 Background Theory

Pneumatic systems differ from hydraulic systems in that they use a compressible fluid and hydraulics do not. Due to the nature of compressible fluids in hydraulic systems, fluid temperature is an important parameter. If a gas such as air gets too hot it can permanently damage the system.

Pascal's Law

Pascal's law says that pressure is equal to force per unit area as shown in equation (1). Therefore, by multiplying the pressure of the fluid by the area of the site of contact between the fluid and actuator, we can calculate the force applied to the actuator.

$$P = \frac{F}{A} \quad (1)$$

3.0 Simscape

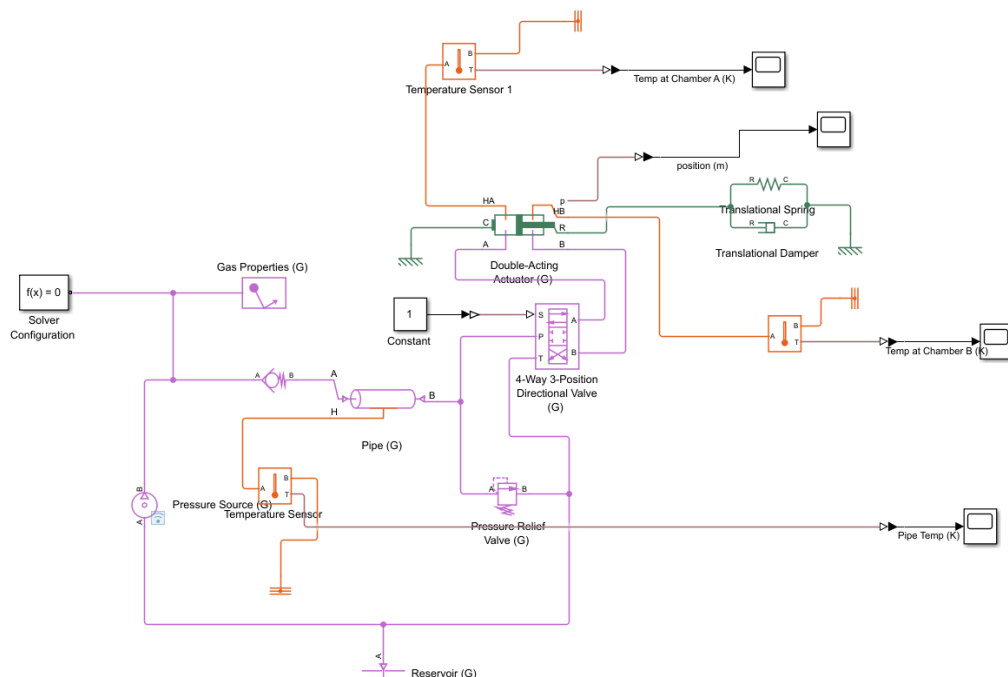


Fig.1 Simscape model

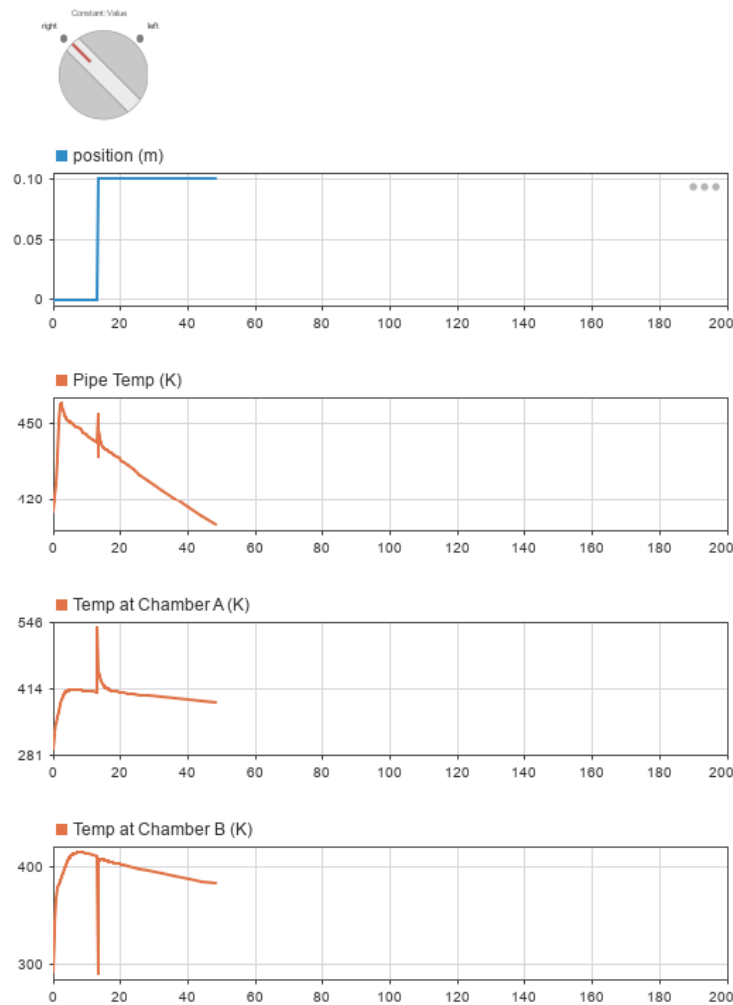


Fig. 2 Instrumentation Panel

Simscape Components

pressure source - creates a pressure differential within the fluid to draw away from the reservoir

Pipe - used to model how pressure and temp change within a pipe when the outlet is choked

check valve - creates a one way flow in pneumatic system

pressure relief valve - releases excess pressure within the system

4-way 3-position directional control valve - controls pressure and direction of flow for 4 ports, with three different flow possibilities

double-acting actuator- generates translational mechanical motion from pressurized gas

Thermal mass - simulates mass properties of pneumatic fluid

temperature measurement block - measures temperature of pneumatic fluid

gas properties block - creates properties of pneumatic fluid

mass-spring-damper system - generates dynamic response to actuator forces

rotary switch - controls the direction of fluid flow

dashboard scopes.- generates plots of thermal and mechanical responses over time

4.0 Results

The graphs generated by the Simscape model show the piston displacement as a 0th order system. In other words, it instantaneously moves from full extension to retraction. While this is not possible in reality, it is a good estimation of an extremely fast cycle piston.

The temperature measurements gathered by the simulation show a 1st order system response where the temperature quickly climbs to around 460 K and then steadily falls back to room temp.

5.0 Instrumentation Panel

The purpose of an instrumentation panel such as the one in Fig. 2 is to show the instantaneous temperature of the pneumatic fluid or piston position. This allows the user to simulate where and when the temperature of the fluid dips below a minimum or how piston position affects temperature of the system in real time.

6.0 Discussion

This lab allowed the simulation of mechanical piston movement from a pneumatic system. While the results represent a very ideal case, it is still important to recognize the usefulness of having a working simulation to predict failure of a possible system. Additionally it is worth noting that the temperature of the pneumatic fluid slowly returns to room temperature due to the simulation accounting for heat escaping through the walls of the pipe.

7.0 Conclusion

Overall, this lab was a good introduction to the instrumentation panel and is a good method of showing the importance of real time measurement. Whereas the Simscape model itself helps the lab user to better understand the necessary components of a pneumatic system and how that system translates to mechanical motion applications.